

## SESSION 3 - Advanced Mobile Testing (APK, IPA) + Real Devices vs Emulators

**1. Download any free APK file (for example, a simple calculator app) and install it on your Android emulator using Android Studio's Device Manager.**

**Ans:** •Downloaded a free Calculator APK file.

- Opened Android Studio Device Manager and started the Android Emulator.
- Installed the APK by dragging and dropping it onto the emulator.
- Waited for the installation process to complete.
- Launched the application from the emulator.

**2. Open the installed app on your emulator, rotate the device from portrait to landscape and back, and note any UI issues or layout problems that appear. Use the emulator's rotate button to change orientation and take screenshots if you spot any glitches.**

**Ans:** • UI Issues Observed: NO UI issues observed.

- Layout Behavior: The layout adjusted correctly when switching from portrait to landscape and back to portrait mode.

**3. Simulate turning off the internet connection in your emulator, then try to use a feature in the app that requires network (like fetching data or logging in). Record the error message or behavior you observe.**

**Ans:** Error Message Observed:

“No Internet Connection”

- Behavior Observed:

The app was unable to load data and the login request failed until the internet connection was restored.

**4. Test the login or registration flow of any installed app on both your emulator and a real Android phone (if available). Compare the experience and list at least two differences you notice in speed, UI, or notifications.**

**Ans:**

| Features       | Emulator        | Real Device    |
|----------------|-----------------|----------------|
| Login Speed    | Slightly slower | Faster         |
| UI Performance | Less smooth     | Smoother       |
| Notification   | Limited support | Works properly |

**5. Pick any app you use daily (Instagram, Zomato, Paytm, etc.) and describe how you would test push notifications using both an emulator and a real device. Mention at least one limitation of emulators for this test.**

**Ans:** Instagram:

Using Instagram:

- Install and login to Instagram on both emulator and real device.
- Send a message from another account to trigger a notification.

- Verify that the notification appears.
- Tap the notification and verify that the correct chat screen opens.

Limitation of Emulator:

- Notification sound and vibration cannot be tested accurately on an emulator.